

TRƯỜNG ĐẠI HỌC CÔNG NGHIỆP TP.HCM

Khoa Công Nghệ Điện Tử - Bộ môn Điện Tử Máy Tính

BÀI GIẢNG

**HỆ THỐNG NHÚNG
(Embedded System)**

TP.HCM-2024

NỘI DUNG MÔN HỌC

Chương 1: GIỚI THIỆU HỆ THỐNG NHÚNG

Chương 2: KỸ THUẬT GIAO TIẾP NGOẠI VI

Chương 3: CẤU TRÚC PHẦN MỀM

Chương 4: THIẾT KẾ HỆ THỐNG NHÚNG

CHƯƠNG 2: KỸ THUẬT GIAO TIẾP NGOẠI VI



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2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP
3. MỘT SỐ KỸ THUẬT GIAO TIẾP CƠ BẢN TRÊN HTN

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

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2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.1 Introduction

- + Serial communication uses **one** or **two transmission lines** to send and receive data, and that data is continuously sent and received **one bit** at a time.
- + A **benefit** of serial communications is **low pin counts**. Serial communications can be performed **with just one I/O pin**, compared to eight or more for parallel communications.
- + For embedded systems that must interface with a general-purpose computer, a serial interface is often **easy to use**.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.2 Serial communication terminology

In serial communication, data is sent **one bit at a time** using **one signal line**, so in order for the receiving side to accurately receive the data, the sending side must know at what speed it is sending each bit.

❑ Synchronous communication

- + This method sends and receives data **synchronized to a clock** generated by the other equipment or by a self-generated clock.
- + Communication is performed is based on a synchronization signal added to each bit from the sending side.
- *Efficiency* but the **transmission procedure becomes complicated**.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.2 Serial communication terminology

For normal communication, both the sending side and the receiving side initially agree on how many bits to transfer each second. Then, each creates a **synchronization signal of a frequency** that matches that transmission rate.

❑ Asynchronous communication

- + This method sends and receives data synchronized to each side's **own self-generated clock**.
- + In this method, data is sent and received **one bit at a time on one data line**. Therefore, each side's **communication condition settings** *must be initially matched*.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.2 Serial communication terminology

□ Transmission rate

+ Transmission rate specifies the **number of bits** to send **each second**. The unit is **bps** (bits per second) and is selected from 300, 600, 1200, 2400, 4800, 9600, 19200, and so on. By matching the settings and timing, **the data delimiters correspond**, and data can be normally sent and received.

→ A **start bit** is added to each item of *data frame* to acquire the correct timing, and a **stop bit** indicates the end of the *data frame*.

→ A **parity bit** of "1" or "0" is added to the data so as to make the number of "1" data bits even for EVEN and odd for ODD.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.2 Serial communication terminology

❑ Full-duplex communication

+ A method where send and receive both have their own transmission line so data can be simultaneously sent and received.

❑ Half-duplex communication

+ A method where communication is performed using one transmission line while switching between send and receive. Therefore, data can be sent or received, but not at the same time.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.2 Serial communication terminology

□ Handshake (flow control)

+ The handshake (flow control) is a function that maintains the reliability of communications. Specifically, data can be transferred while each side checks the sending and receiving of data.

- A signal is sent from the sending side to the receiving side that states, "data is being sent".
- And the receiving side receives that signal and reads the data from the signal line. Then it sends a reply to the sending side that states, "the data was received".

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.2 Serial communication terminology

□ Master/slave

- + This describes a bus where **one device is the master** and **others are slaves**.
- + Master/slave buses are usually synchronous, as the master often supplies the timing clock for data being sent along in both directions.

□ Multi-master

- + A multi-master bus is a master/slave bus having **more than one master**.
- + Multi-master buses must have **an arbitration scheme** can settle conflicts when **more than one master** wants to control the bus at the same time.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.2 Serial communication terminology

❑ Point-to-point or Peer interfaces

+ They are where two devices have a peer relation to each other, there are no masters or slaves. Peer interfaces are most often asynchronous.

❑ Multi-drop

+ An interface in which there are several receivers and one transmitter.

❑ Multi-point

+ A bus in which there are more than two peer transceivers. This is different from a multi-drop interface as it allows bidirectional communication over the same set of wires.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.3 Types of serial communication protocols

1) RS-232

- + TIA/EIA-232-F (typically referred to as **RS-232**) is a common interface that can be found on almost every personal computer or embedded system.
- + An RS-232 bus is an **unbalanced bus** capable of **full-duplex** communication between two receiver/transmitter pairs, named data terminal equipment (DTE) and data communication equipment (DCE). It has only the capable of speeds up to **20Kbps** and distances up to **15m**.
- + Almost every microcontroller vendor has products that include **hardware support for RS-232**, called **UARTs**.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.3 Types of serial communication protocols

□ *RS-422 and RS-485*

- + TIA/EIA-422-B (typically referred to as **RS-422**) and TIA/EIA-485-A (typically referred to as **RS-485**) are **balanced**, twisted-pair interfaces capable of speeds up to **10Mbps** and distances up to **1200m**.
- + Being **differential** buses, each uses signals from **1.5V to 6V** to transmit the data.
- With a **differential, balanced bus**, **noise immunity is increased** over a comparable *single-ended, unbalanced bus* such as *RS-232*.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.3 Types of serial communication protocols

2) RS-422

+ The RS-422 interface is a **multi-drop** interface, giving unidirectional communication over a pair of wires from one transmitter to several receivers, up to **10 unit loads (UL)**.

+ If the devices receiving the data **wish to communicate back** to the transmitter, the designer must use *a separate, dedicated bus* between each receiver and the transmitter.

→ Using *this return bus* will allow **full-duplex** transmissions. For that reason, RS-422 **is seldom used** between **more than two nodes**.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

2.1.3 Types of serial communication protocols

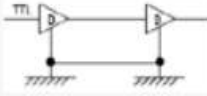
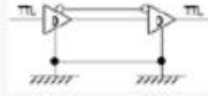
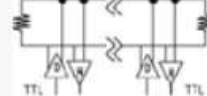
3) RS-485

- + The RS-485 interface, on the other hand, is a bidirectional communication over one pair of wires between several transceivers. The specification states that the bus can include up to **32 unit loads (UL)** worth of transceivers.
- + Many manufacturers produce fractional-UL transceivers, thereby increasing the maximum number of devices to well **over 100 UL**.
- + In fact, several converters exist to go from RS-232 to RS-485 and back, however, that RS-232 is a **full-duplex** interface while RS-485 is **half-duplex**.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

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2.1 Sơ

Parameter	RS-232C	RS-422A	RS-485
Transmission mode	Simplex	Multi-point simplex	Multi-point multiplex
Max. connected devices	1 driver 1 receiver	1 driver 10 receivers	32 drivers 32 receivers
Max. transmission rate	20Kbps	10Mbps	10Mbps
Max. cable length	15m	1200m	1200m
Operation mode	Single-ended (unbalanced type)	Differential (balanced type)	Differential (balanced type)
Connection image			
Features	Short distance Full-duplex 1:1 connection	Long distance Full-duplex, half-duplex 1:N connection	Long distance Full-duplex, half-duplex N:N connection

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

4) I2C

- + The Inter-Integrated Circuit (I2C) is developed by **Philips Semiconductors**.
- + The I2C bus is a **half-duplex, synchronous, multi-master** bus requiring only **two signal wires**: data (SDA) and clock (SCL).
- + I2C uses an **addressable communications** protocol that allows the master to communicate with individual slaves using a 7-bit or 10-bit address (rare).
- + During communication with slave devices, the master generates all clock signals for both communication to and from the slave. Each communication begins with the master generating a start condition, an 8-bit data word, an acknowledge bit, followed by a stop condition or a repeated start.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

5) SPI

- + The Serial Peripheral Interface (SPI) is a synchronous serial bus developed by **Motorola**.
- + The SPI bus consists of **four signals**: master out slave in (MOSI), master in slave out (MISO), serial clock (SCK), and active-low slave select (/SS).
- + As a **multi-master/slave protocol**, communications between the master and selected slave use the **unidirectional MISO and MOSI lines**, to achieve data rates over 1Mbps in full **duplex** mode.
- + The data is clocked simultaneously into the slave and master based on SCK pulses the master supplies.

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2.1 Serial Communication Protocols

6) Microwire

- + Microwire is a **three-wire synchronous** interface (**National Semiconductor**).
- + Similar to SPI, Microwire is a **master/slave bus**, with serial data out of the master (SO), and serial data in to the master (SI), and signal clock (SK), correspond to SPI's MOSI, MISO, and SCK, respectively. Besides, there is also a chip select signal, which acts similarly to SPI's /SS.
- + As **full-duplex** bus, Microwire is capable of speeds of **625Kbps** and faster.
- + Microwire devices from National come with different protocols, based on their data needs → Microwire permits variable length data, and also specifies a “continuous” bitstream mode.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.1 Serial Communication Protocols

7) 1-Wire

- + 1-Wire bus is an **asynchronous** (developed by **Dallas Semiconductor**), master/slave bus with **no protocol** for **multi-master**.
- + Like the I2C bus, 1-Wire is **half-duplex**, using an open-drain topology on a single wire for bidirectional data transfer.
- + However, the 1-Wire bus also allows the data wire to transfer power to the slave devices, although this is somewhat limited.
- Though limited to a maximum speed of **16Kbps**, bus length can be upwards of **300m**, given the **proper pull-up resistor**.

2. CÁC LOẠI PHƯƠNG THỨC GIAO TIẾP

2.2 Parallel Communication

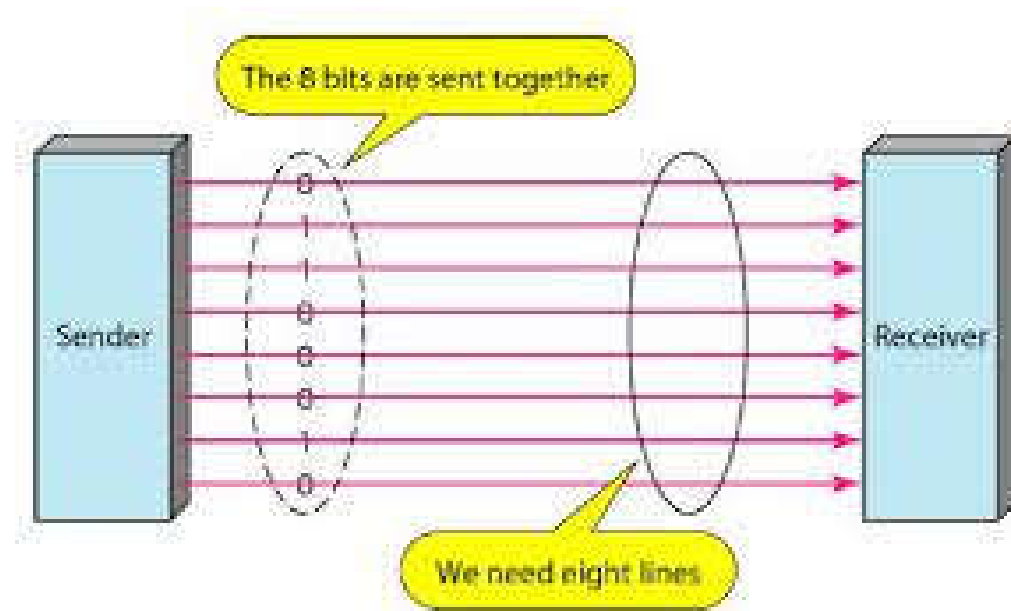
- A parallel interface refers to a multiline channel, with each line capable of transmitting **several bits of data simultaneously**.
- In parallel transmission, multiple bits (usually 8 bits / byte character) are sent **simultaneously** on different channels (wires, frequency channels) within the same cable or radio path, and synchronized to a clock.
- Parallel devices have a **wide data bus**: transfer data in words of one or more bytes → there is speed up in parallel transmission bit rate.
- This speed up is a tradeoff versus cost (multiple wires cost more than a single wire, and a parallel cable gets longer)
- The synchronization timing becomes more sensitive to distance.

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2.2 Parallel Communication

Parallel transmission is used when

- A large amount of data is being sent
- The data being sent is time-sensitive
- The data needs to be sent quickly



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